

Divij Dudeja

4th Year Undergraduate
Department of Computer Science (Specialization in Data Science), IIIT Nagpur

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ACADEMIC QUALIFICATIONS

Year	Degree/Certificate	Institute	GPA/%
2022 – Present	B.Tech	Indian Institute of Information Technology Nagpur	8.20/10
2022	Class XII (CBSE)	Modern Delhi Public School, Faridabad	87.6%
2020	Class X (CBSE)	Modern Delhi Public School, Faridabad	90.6%

SCHOLASTIC ACHIEVEMENTS

- Ranked among the top 200 **NTSE Stage I** qualifiers in Haryana, earning merit-based scholarship for academic excellence. 2020
- Achieved International Ranks 57 and 157 in **NCO**, along with state distinction medals in **NSO** and **IMO**, conducted by **SOF**. 2018,2019;2017
- Acquired specialist rating (Max- 1402) in **Codeforces**, and rated 3 Star (Max- 1710) in **CodeChef**. 2024
- Runner up** in Analytica a 24hr Data Science Hackathon Conducted at IIIT Nagpur. 2024
- Special Mention** in Marketwise, Hackathon by E-Cell IIIT Nagpur. 2025

WORK EXPERIENCE

AI Engineer Intern | EmployWise |Remote April'26 – Present

- Shipped an N8N automation pipeline that eliminated 45 hrs/week of manual cross-team audit work, auto-routing mismatches to responsible owners.
- Developing an internal MCP (Model Context Protocol) server to standardize and centralize AI tool access across the organization.

Data Science Internship | Asea Brown Boveri (ABB) |Hyderabad (On-Site) June'25 – Dec'25

- Filed a **PATENT** for building an end-to-end **AI pipeline** with the use of **ML + CV** models for optimizing solar plant profitability through: integrating weather-driven energy forecasting (P10/P50/P90) and with panel-level defect detection and health scoring from images, and a **PPO-based reinforcement learning model** to optimize electricity trading decisions and accurately estimate real-world solar plant output.
- Wrote a **research paper** on **SMART** a 45.51M parameter Small Language Model (SLM), using **Tree-LSTM + MANN + Transformer** architecture, to deliver precise information from engineering document manuals. [Link](#)
- Built a RAG-based document automation tool using**LangChain + Gemini** that reduced manual RFQ processing time by 40%.
- Developed a graph-driven resilience assessment framework for distribution networks to simulate contingency scenarios and evaluate survivability. Worked on an RL based optimization model, integrating a cost-aware ranking mechanism to balance resilience performance against operational penalties.

Software Developer Internship | Pranav Vikas Pvt. Ltd |Faridabad (On-Site) May'24 – June'24

- Engineered and optimized the **Battery Cooling System** software, translated client requirements for **Switch Mobility** Electric Sathi vehicle and build the software with a control logic for a shared cabin and battery cooling architecture for the AC system.

Data Scientist Intern | Ecocat India Pvt Ltd |Faridabad (On-Site) Dec'23 – Jan'24

- Performed **EDA** and applied **PCA** for dimensionality reduction on manufacturing process and emission-test data to improve feature representation and support anomaly detection and classification models.
- Developed classification model using **Logistic Regression** and anomaly detection using **Isolation Forest** to identify abnormal batches and process deviations in catalyst manufacturing, enabling early corrective actions.

PROJECTS

Blog Writing AI Agent | LangGraph, LangChain, Groq, Gemini, Python, Streamlit Feb'26 – Mar'26

- Built a **LangGraph**-based blog writing agent that automates end-to-end content generation through a structured workflow.
- Implemented **adaptive routing** to decide if a topic needs **research** before generation, enabling closed-book, hybrid and open-book blog creation.
- Designed parallel worker-based task execution with reducer-based merging to synthesize section-level outputs into a coherent final blog draft.
- Added **research-grounded** and **image-aware generation** using Tavily-backed evidence collection and automated image planning with placeholders and graceful fallback handling.

Aarogyam | Langchain, Groq, LLM, Automation, VLMs, Neo4j GraphDB, Flask, TensorFlow Jan'25 – Mar'25

- Built a healthcare web application with **disease prediction systems** using TensorFlow and Scikit-learn, achieving **95%+ risk prediction accuracy**.
- Integrated **knowledge graph-based chatbots** and OCR-enabled medical report management.
- Built a **hospital finder** using HERE API and a **diet chatbot** with Groq API, enabling users to locate nearby hospitals and receive personalized dietary advice with 95% accuracy.

TECHNICAL SKILLS

- Languages** : C, C++, Python, SQL, Fast API, HTML, CSS, JavaScript, Docker
- AI & ML Stack** : TensorFlow, PyTorch, Keras, Scikit-learn, Pandas, Numpy, OpenCV, LangChain, LangGraph, MCP, NLP
- Generative AI** : Transformers, Prompt Engineering, LORA/QLoRA, Agentic AI, Pydantic, Vector DB, N8N
- Tools** : Chatgpt, Claude Code, Ollama, Hugging Face, Jupyter, VS Code, Anaconda Navigator, Google Colab, Power BI, Github, Cursor, MS Excel, Graphify

EXTRA CURRICULAR ACTIVITIES

- Won the **District Tournament** in Men's Throw ball. 2019
- Public Speaking: Anchored in events including **Abhivyakti** , **TantraFiesta**, and E-cell events. 2023,2024
- Invited as a Judge for debate competitions at **Infacto**, evaluating participants on content, delivery, and argumentation. 2025
- Secured 3rd position in Intra-School **Model United Nations (MUN)**. 2018

RELEVANT COURSES

Machine Learning	Artificial Intelligence	Deep Learning
Natural Language Processing	Operating Systems	Computer Vision
Database Management System	Data Structures	Intro to Generative AI

POSITION OF RESPONSIBILITY

- Head of Content and Anchoring Team, Tantrafiesta (Technical Fest)** (July'24 – Nov'24) — Led a team of 20+ anchors and content curators for spearheading end-to-end content strategy, coordination, and on-stage anchoring for the institute's tech fest.
- Head of Data Science and AI , CRISPR Club** (Jul'23 – June'24) — Conducted a national level AI+ML 24 hour hackathon (Solvathon 3.0) at the technical fest (TantraFiesta).
- Head of Marketing, Orator Club** (Jul'23 – June'24) — Led marketing initiatives, driving sponsorship acquisition and expanding MUN's & Infacto's outreach across multiple colleges in Nagpur and online social media campaigns to all colleges in India.